

- 1 ABC is a straight line and BCD is a triangle.

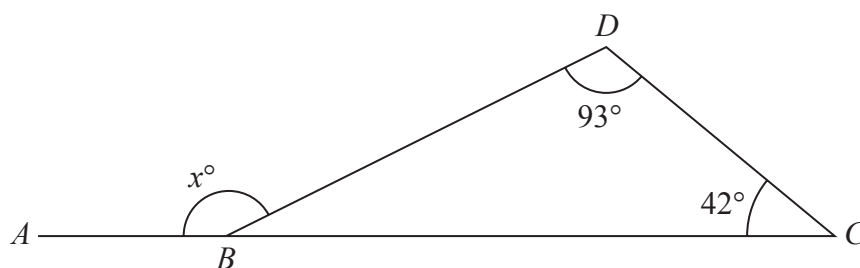


Diagram **NOT**
accurately drawn

- (a) Work out the value of x

$$x = \dots\dots\dots$$

(2)

PO , RO , SO and TO are four straight lines.

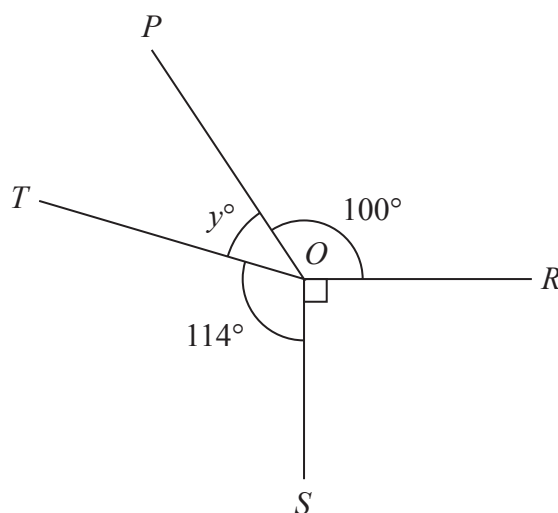


Diagram **NOT**
accurately drawn

- (b) (i) Work out the value of y

$$y = \dots\dots\dots$$

(2)

- (ii) Give a reason for your answer.

(1)

(Total for Question 1 is 5 marks)

2 The diagram shows triangle ABD

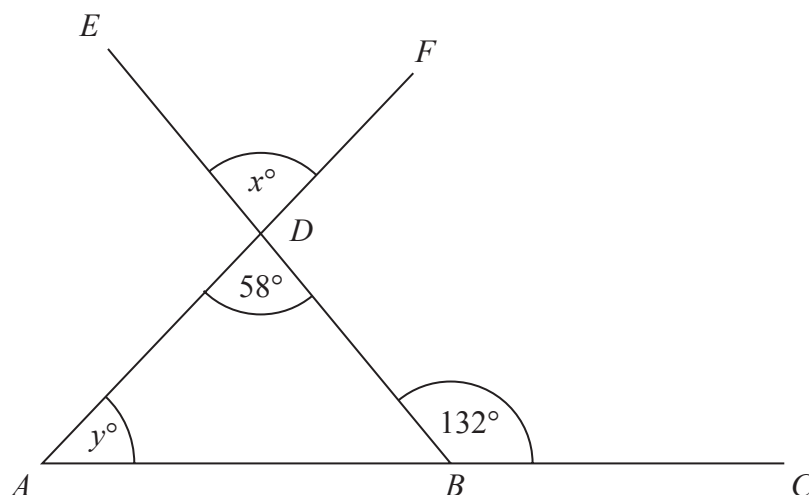


Diagram **NOT**
accurately drawn

ABC , BDE and ADF are straight lines.

angle $CBD = 132^\circ$ angle $ADB = 58^\circ$

(a) (i) Write down the value of x

$x =$

(ii) Give a reason for your answer.

(2)

(b) Work out the value of y

$y =$

(2)

(Total for Question 2 is 4 marks)

3

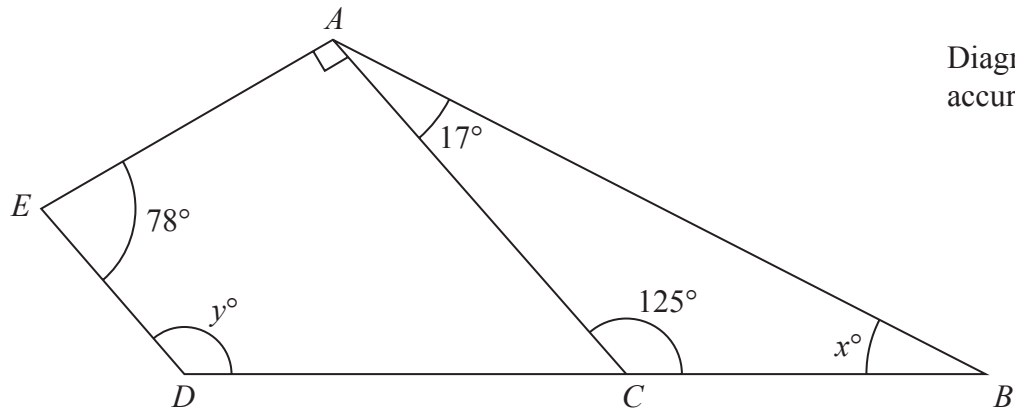


Diagram **NOT**
accurately drawn

$ABDE$ is a quadrilateral.
 ABC is a triangle.
 DCB is a straight line.

(a) (i) Work out the value of x .

$x = \dots\dots\dots$
 (1)

(ii) Give a reason for your answer.

(1)

(b) Work out the value of y .
 Give a reason for each stage of your working.

$y = \dots\dots\dots$
 (3)

(Total for Question 3 is 5 marks)

4

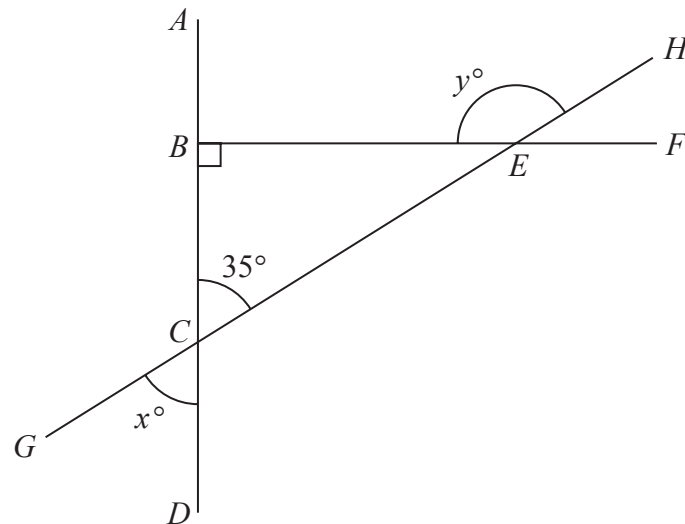


Diagram **NOT**
accurately drawn

In the diagram, BCE is a right-angled triangle.
 $ABCD$, BEF and $GCEH$ are straight lines.

Angle $BCE = 35^\circ$

(a) (i) Find the value of x

$x = \dots\dots\dots$
(1)

(ii) Give a reason for your answer.

(1)

(b) (i) Work out the value of y

$y = \dots\dots\dots$
(2)

(ii) Give a reason for your answer.

(1)

(Total for Question 4 is 5 marks)

5 The diagram shows two parallel lines AB and DEF

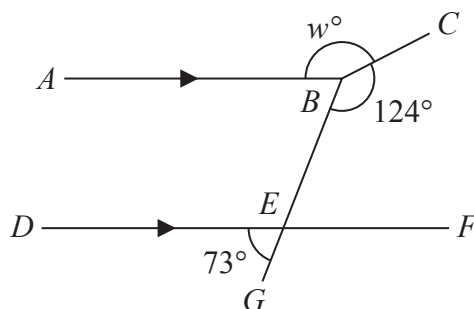


Diagram **NOT**
accurately drawn

BEG is a straight line.

$$\text{angle } DEG = 73^\circ \quad \text{angle } EBC = 124^\circ \quad \text{angle } ABC = w^\circ$$

Work out the value of w

Give reasons for each stage of your working.

$$w = \dots\dots\dots$$

(Total for Question 5 is 4 marks)

6

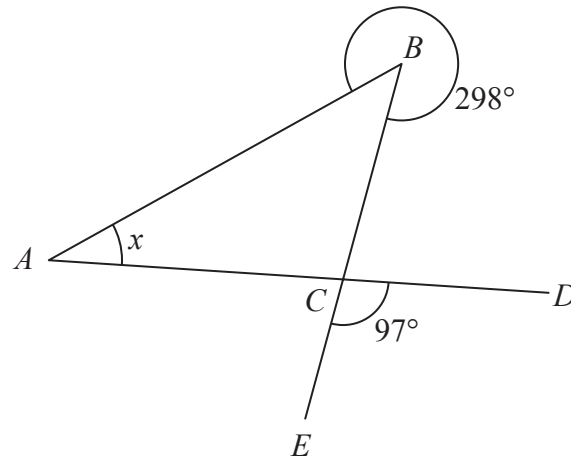


Diagram **NOT**
accurately drawn

ABC is a triangle.

D and E are points such that ACD and BCE are straight lines.

reflex angle $ABC = 298^\circ$

angle $ECD = 97^\circ$

Work out the size of angle x .

Give a reason for each stage of your working.

$x = \dots\dots\dots^\circ$

(Total for Question 6 is 4 marks)

7

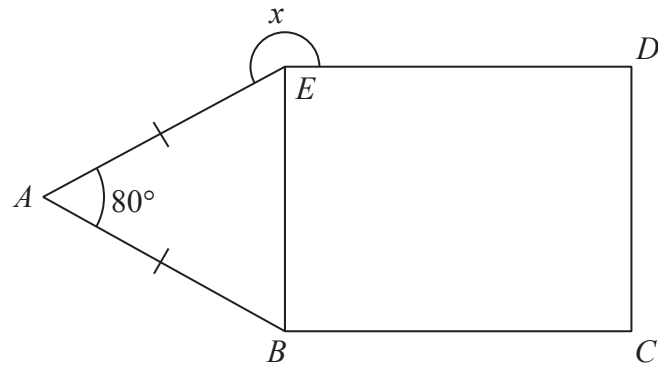


Diagram **NOT**
accurately drawn

$BCDE$ is a rectangle.

ABE is an isosceles triangle.

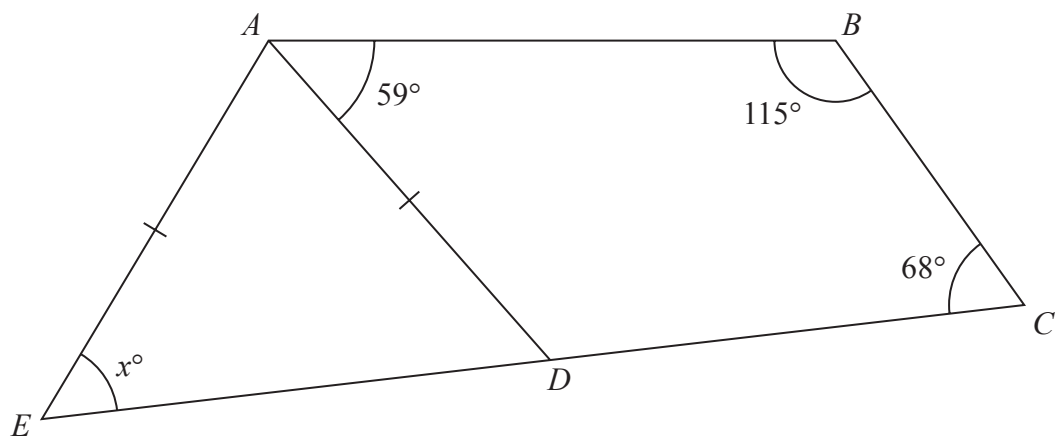
$AB = AE$

Angle $BAE = 80^\circ$

Work out the size of angle x .

(Total for Question 7 is 3 marks)

- 8 The diagram shows quadrilateral $ABCD$ and isosceles triangle ADE , where $AE = AD$.



EDC is a straight line.

Work out the value of x .

Give a reason for each stage of your working.

$x = \dots\dots\dots$

(Total for Question 8 is 4 marks)

- 9 The diagram shows a trapezium $ABCD$ in which AB and DC are parallel.

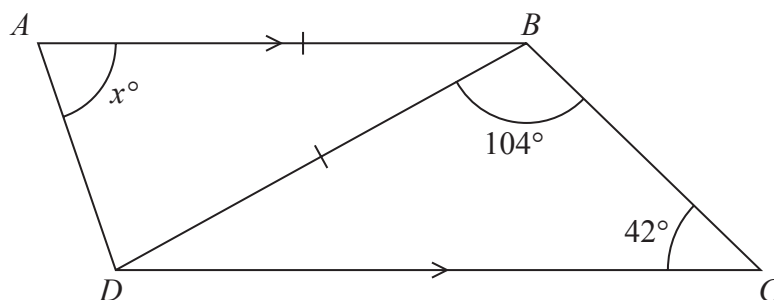


Diagram **NOT**
accurately drawn

$$AB = DB$$

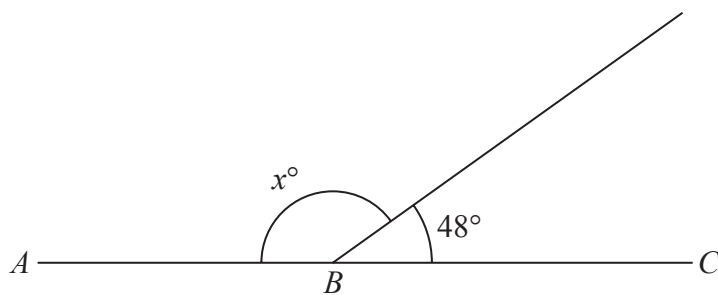
Work out the value of x .
Give a reason for each stage of your working.

$$x = \dots\dots\dots$$

(Total for Question 9 is 4 marks)

10

Diagram **NOT**
accurately drawn



ABC is a straight line.

(a)(i) Work out the value of x

$x =$
(1)

(ii) Give a reason for your answer to (i)

.....
(1)

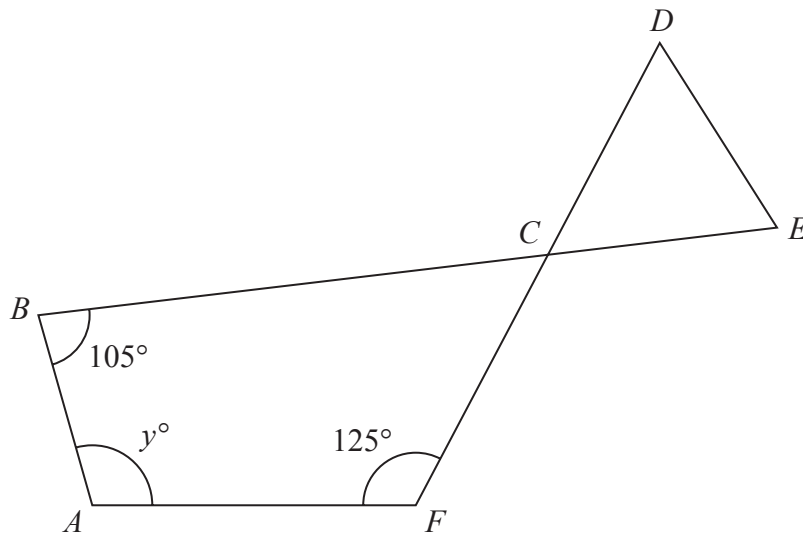


Diagram **NOT**
accurately drawn

CDE is an equilateral triangle.

$ABCF$ is a quadrilateral.

BCE and DCF are straight lines.

- (b) Work out the value of y
You must show your working.

$y = \dots\dots\dots$
(3)

(Total for Question 10 is 5 marks)

11

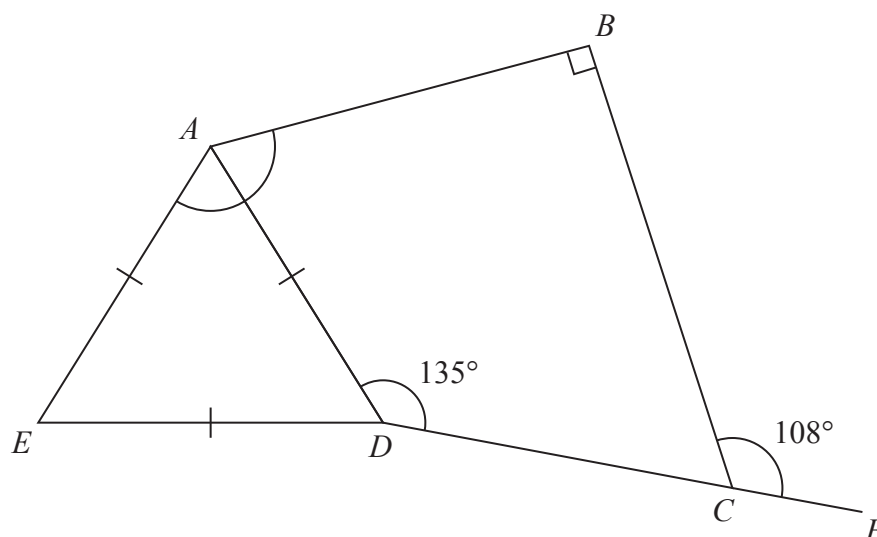


Diagram **NOT**
accurately drawn

$ABCD$ is a quadrilateral.

ADE is an equilateral triangle.

DCF is a straight line.

Work out the size of angle EAB .

Give a reason for each stage of your working.

(Total for Question 11 is 5 marks)

12 The diagram shows two triangles, CDB and BDA .

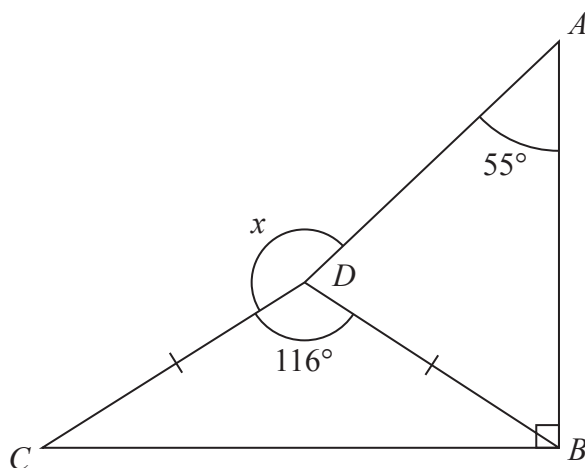


Diagram **NOT**
accurately drawn

$$DC = DB$$

Angle $ABC = 90^\circ$

Angle $CDB = 116^\circ$

Angle $DAB = 55^\circ$

Work out the size of the angle marked x .

Give a reason for each stage of your working.

(Total for Question 12 is 5 marks)

13

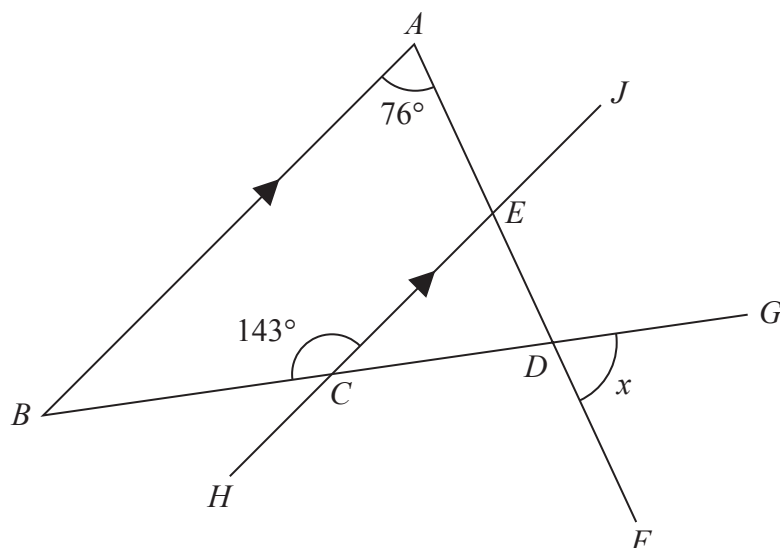


Diagram **NOT**
accurately drawn

ABD is a triangle.

$AEDF$, $BCDG$ and $HCEJ$ are straight lines.

BA is parallel to $HCEJ$.

Work out the size of the angle marked x .

(Total for Question 13 is 3 marks)

14

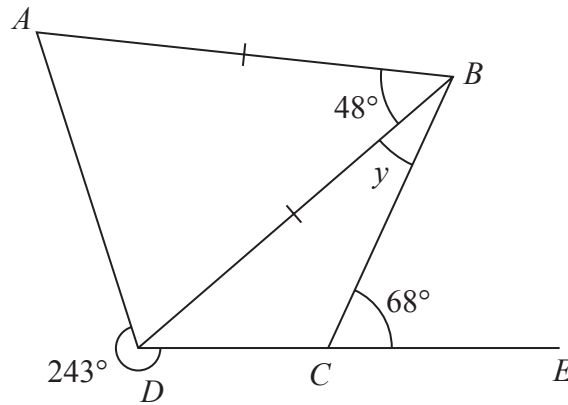


Diagram **NOT**
accurately drawn

ABD is an isosceles triangle with $AB = DB$.
 DCE is a straight line.

Angle $ABD = 48^\circ$

Angle $BCE = 68^\circ$

Reflex angle $ADC = 243^\circ$

Work out the size of the angle marked y .
Give a reason for each stage in your working.

(Total for Question 14 is 5 marks)

15

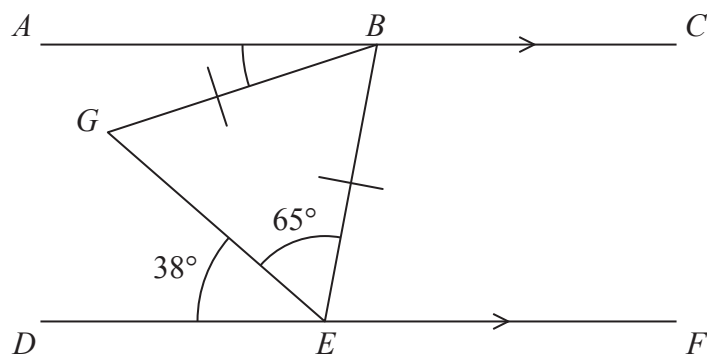


Diagram **NOT**
accurately drawn

ABC and DEF are parallel lines.

$BG = BE$

Angle $DEG = 38^\circ$

Angle $GEB = 65^\circ$

Find the size of angle ABG .

(Total for Question 15 is 3 marks)